Government College Of Engineering Aurangabad

(An Autonomous Institute Of Government Of Maharashtra)

SY (CBCS) Class Test - 1

CSPCC2003 :Operating System

Time: One Hour | Date of Examination : 04/09/2024 | Max. Marks: 15

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. ***Question 1 is compulsary.***
2. ***Attempt any two from Q2,Q3,Q4***
3. ***Assume suitable data if necessary and state it clearly***
4. ***Use of non-programmable calculator is allowed***

Q. No.	Questions	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q.1.	Find average TAT & average WT for a given process by using priority scheduling with non-preemptive.	CO2	BT3	05

Process ID	Arrival Time (in ms)	CPU Time Needed(in ms)	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5

Note : higher priority represent higher number.

Q.2	Write a short note on 1. System call 2. Buffering 3. Spooling 4. User interface 5. Command interpreter	CO1	BT1	05
Q 3.	What do you understand by multiprocessing operating system and Explain symmetric and multicore operating system.	CO1	BT2	05
Q.4	Discuss the process control block and interprocess communication	CO2	BT2	05

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SY (CBCS) Class Test - 2

CSPCC2003 :Operating System

Time: One Hour | Date of Examination : 24/10/2024 | Max. Marks: 15

"Verify the course code and check whether you have got the correct question paper"

N.B:-

1. ***Attempt any three questions.***
2. ***Assume suitable data if necessary and state it clearly***
3. ***Use of non-programmable calculator is allowed***

Q. No.	Questions	Course Outcomes (Cos)	Bloom's Taxonomy Levels (BTLs)	Marks
Q 1.	Differentiate between Fixed and variable partitioning. Explain fragmentation.	CO3	BT4	05
Q.2	Enlist the different steps involved in page replacement algorithm with neat diagram and Apply LRU page replacement algorithm on following string . 3 frames 6,1,1,2,0,3,4,6,0,2,1,2,3,2	CO3	BT3	05
Q 3.	Describe SSTF and SCAN algorithms and state the advantages and disadvantages.	CO4	BT2	05
Q.4	Discuss the following concepts. a. Disk Management b. Virtual Memory	CO3/CO4	BT2	05

Government College of Engineering Aurangabad, Chhatrapati Sambhaji Nagar

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End Semester Examination Nov-Dec 2024

Second Year B. Tech. CSE –NEP – Semester – III

CSPCC2003 : Operating System

Time: 2:30Hours 18 DEC 2024 *Max. Marks: 60*

"Verify the Course Code and check whether you have got the Correct Question Paper"

N.B.

1. ***Attempt all questions.***
2. ***Figures to the right indicate full marks.***
3. ***Assume suitable data if necessary and state it clearly.***
4. ***Use of Non-Programmable Calculator and Data Sheet is allowed.***

Q.1	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	State the different types of Operating system. Explain different kinds of services provided by the OS.	*1*	*2*	*06*
B	Justify, Why deadlock situation occur and How to prevent deadlock using prevention techniques.	*2*	*6*	*06*
C	Define the following terms related to OS. 1. Distributed OS 2. Spooling	*1*	*1*	*06*
Q.2	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Illustrate the necessary condition for the critical section problem.	*2*	*3*	*06*
B	Consider the 5 processes and find average TAT using FCFS and SJF primitive algorithms	*2*	*4*	*06*
C	Explain the following terms related to process. 1. Process state 2. Context switch	*2*	*2*	*06*
Q.3	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Illustrate the concept of Demand paging with page table.	*3*	*3*	*06*
B	Enlist the 3 algorithms of dynamic memory allocation and explain.	*3*	*2*	*06*
C	Describe the concept of paging hardware in detail.	*3*	*2*	*06*
Q.4	**Attempt All**	***CO***	***B.T. Level***	***Marks***
A	Write a short note on Disk reliability and stable storage management	*4*	*1*	*06*
B	Analyse SCAN&SSTF disk scheduling algorithms seek time for the following request with the cylinder numbers as 98,183, 37,122,14,124,65,67.The current head position is at 53.	*4*	*4*	*06*
Q.5	**Attempt any two**	***CO***	***B.T. Level***	***Marks***
A	Explain any 3 access mechanisms of file in depth.	*5*	*2*	*06*

process	Arrival time	Burst Time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

B	Enlist and explain any 3 logical structures of directory system.	5	2	06
C	Demonstrate the working of File system for the Android operating system.	5	4	06

Note: CO: Course Outcome; B.T. Level: Bloom's Taxonomy Level